ETH zürich

ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

NextRAN-AI – 05/12/2025

Integrated Systems Laboratory (ETH Zürich)

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PULP Platform
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Next Steps

- **TensorPool:**
  - TeraPool + RedMulE (TensorPool) physical design in 7nm
  - PPA on model microkernels and operators (combined RedMulE&Cores)
- **System Performance:**
  - Simulation speed is impaired by large design size → from RTL simulation to higher abstraction level, e.g. GVSoC
  - TensorPool GVSoC model developement
  - Data-Movement and end-end performance

Programmable Baseband Processors

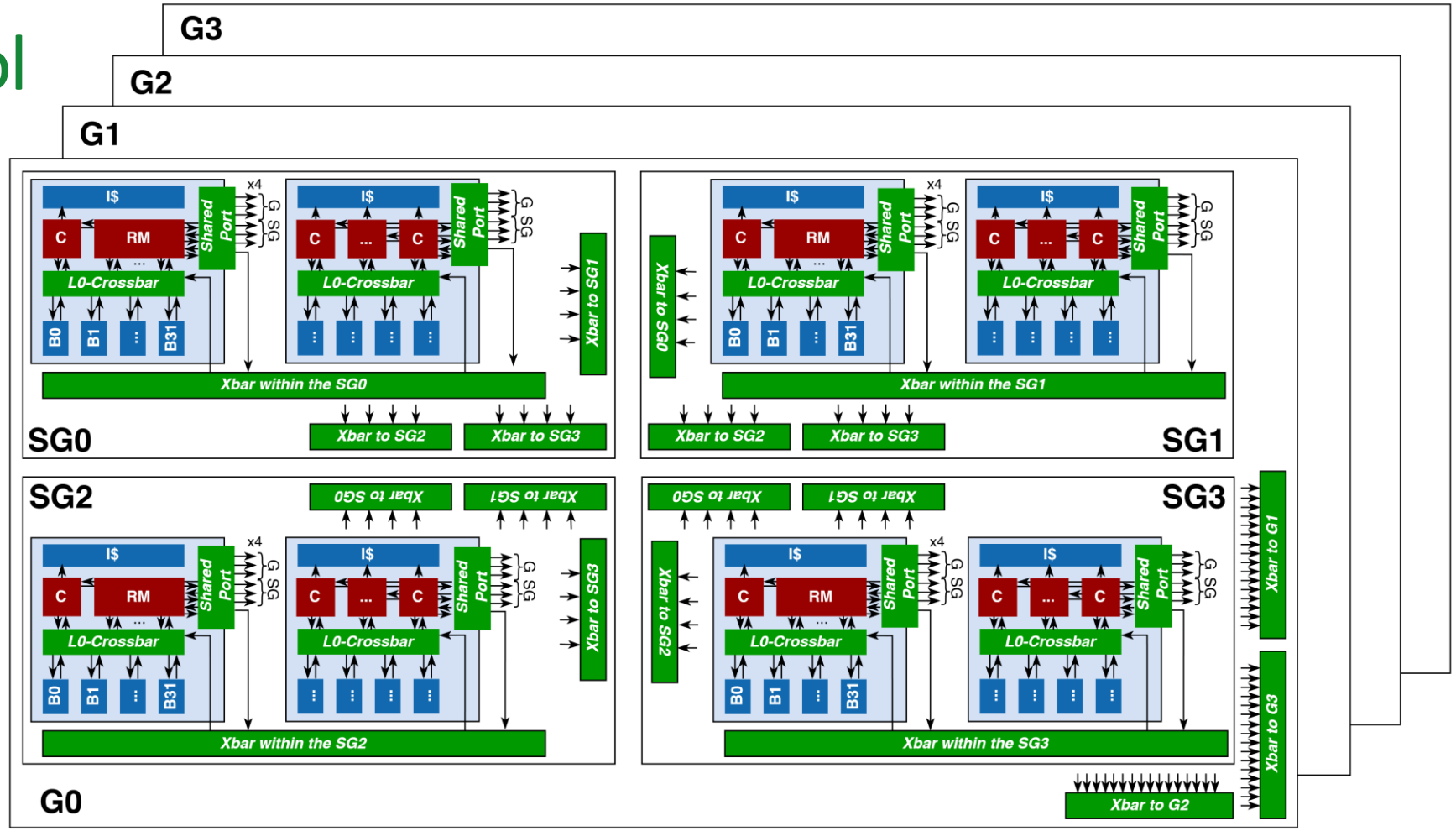
- TeraPool has the lowest power-consumption
- TeraPool has the highest number of PEs sharing the local memory
  - Allows more data-reuse (without workload tiling across memory hierarchies)

	L1 Size	Node	Frequency [GHz]	Performance [TFLOPS@16b]	Power [W]	Area [mm²]	Area&Energy Efficiency [GOPS/W/mm²]
X100 [1]	-	5nm	-	-	35	-	-
Octeon10 [2]	64KiB/PE	5nm	2.5	-	50	-	-
ARC Compact [3]	128KiB/188PEs	5nm	2.5	455.3	72	294	21.5
TeraPool	4MiB/1024PEs	12nm	0.88	3.6	5.5	81.8	8
TeraPool*			2.11	8.64	2.81	27.83	110.5

*Frequency, Power and area normalised to 5nm, multiplying by (12/5), $(5/12)*(0.75/0.8)^2$, $(5/12)^2$

[1] https://www.qualcomm.com/cellular-infrastructures/products/qualcomm-x100-5g-ran-accelerator-card
[2] https://www.marvell.com/products/data-processing-units.html
[3] https://www.nvidia.com/en-us/data-center/l4/

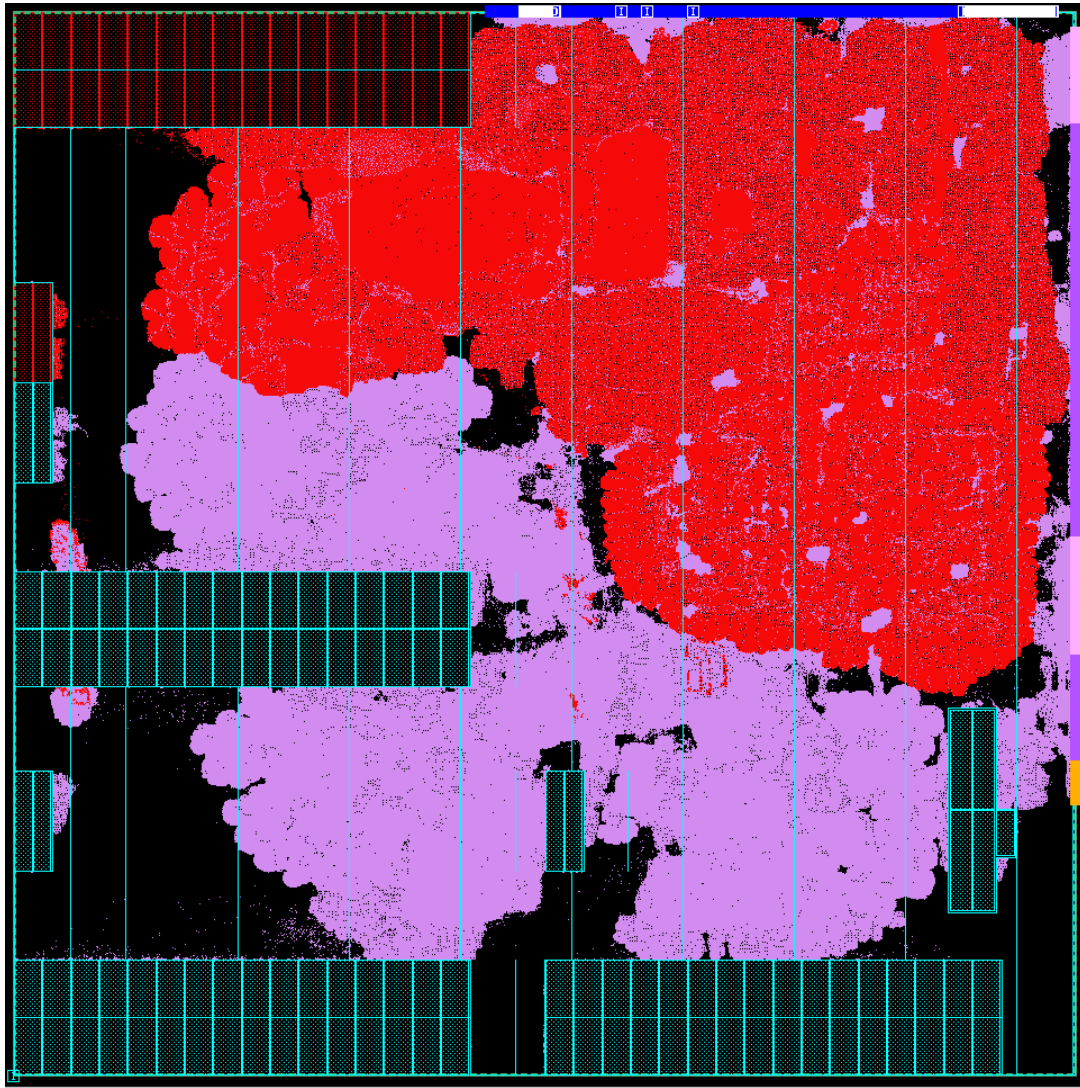
TensorPool



- 4 Tile / SubGroup, 4 SubGroups / Group, 4 Groups / Cluster
- 1 TE / SubGroup, 4PE /Tile → 256 PEs & 16 TEs = 4608 MAC/cycle
- 8x32 TE with 16-ROB entries, grouping factor req 2, response 4

TensorPool SubGroup Floorplan

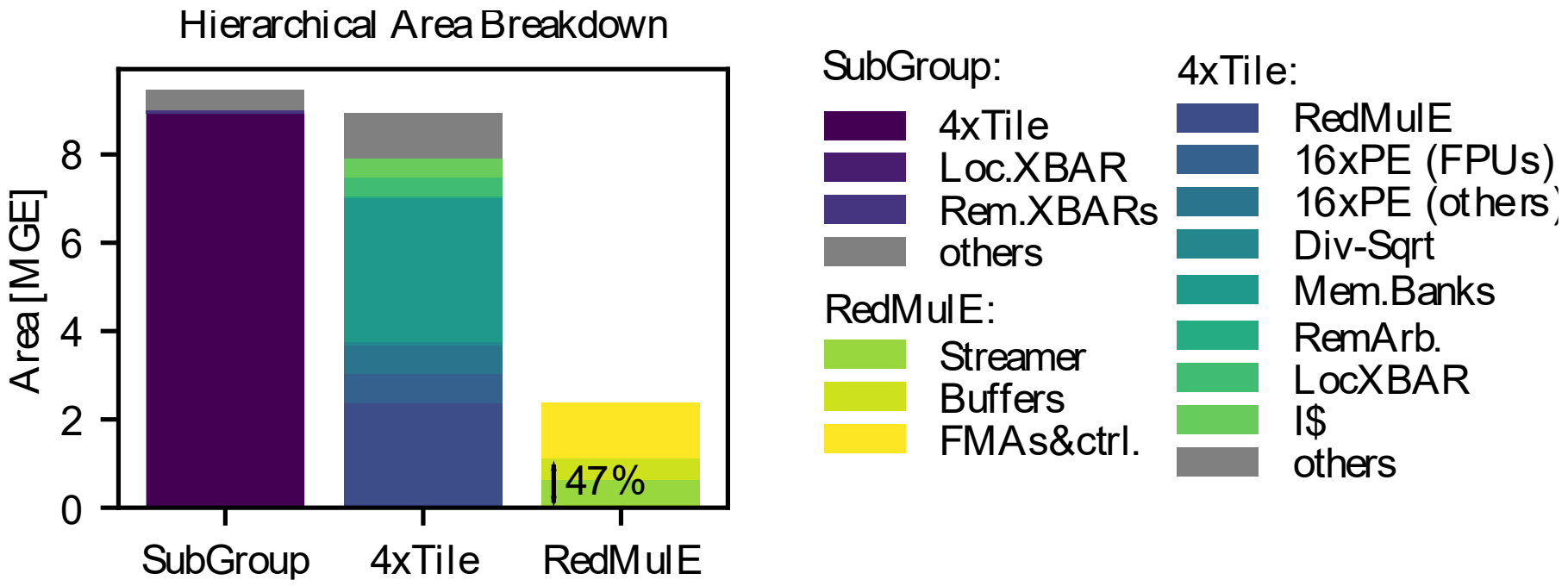
- I leaved more space for the TE
- SubGroups in Groups can be placed with central symmetry as in TeraPool to route the SubGroup to SubGroup connections
- In Group we leave a channel to place Group to Group XBARs off the way



Frequency target (Post Placement)

- I place and route with a 800MHz target, I achieve 900MHz in TT, 25C, 0.75V
- This frequency is aligned to other designs implemented in the same technology and containing the Snitch core

Area Breakdown (Post Placement)



- Compute (FPUs+FMAs) = 20% of SubGroup
- Scratchpad banks = 34% of SubGroup
- Buffers&Streamer = **11.8%** of SubGroup (47% of RedMulE)

Options to reduce the area of the buffers

Buffer	Size (formula for bits)	Size
X_buffer	16-bit*L*2	128B
W_buffer	16-bit*H*H*P	384B
Z_buffer	16-bit*H*L*(P+1)	2KiB
X_ROB	16-bit*ROB_DEPTH*H*(P+1)	1KiB
Y_ROB	16-bit*ROB_DEPTH*H*(P+1)	1KiB
W_ROB	16-bit*ROB_DEPTH*H*(P+1)	1KiB
Z_FIFO	16-bit*L*H*(P+1)	2KiB
TOTAL		**7.7KiB**

It is not 7.7KiB to hide the latency of local scratchpad (64KiB), but the latency of 4MiB -> 2%

- There is on-going work to assess the performance of an outer-product engine.
- This engine fetches the X and W elements on the flight and re-uses the pipeline stages to store accumulation values, reducing the number of buffers.

TensorPool PPA comparison

- I increased compute units by 2.25x (4608 vs 2048 16b-MAC/cycle)
- I kept the flexibility of PEs and added TEs for GEMM workloads

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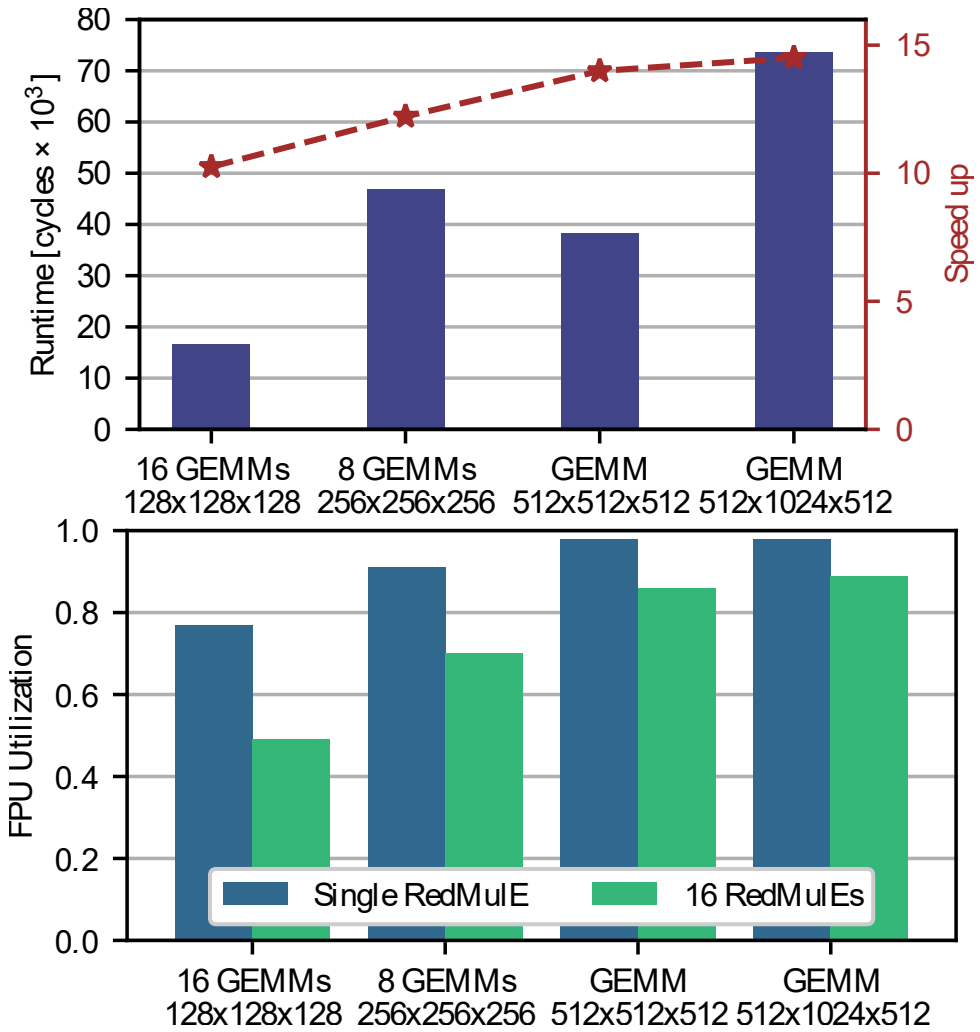
TeraPool	4MiB/1024PEs	12nm	0.88	3.6	5.5	81.8	8
TeraPool*			1.5*	6.17*	2.8*	27.83*	79.18*
TensorPool	4MiB/256PEs+ 16TEs	7nm	0.92	8.3	3	12.32	224.4

Frequency, Power and area normalised to 7nm, multiplying by (12/7), (7/12)(0.75/0.8)², (7/12)²

Domain specialization → better utilization of FMAs



	TeraPool	TensorPool
Throughput [MAC/cycle]	609 of 2048	3643 of 4096
Performance [TFLOPS@16b]	1.8	6.6
Power [W]	3.7	3
Energy Efficiency [TFLOPS@16b/W]	0.5	2.2
Energy&Area Efficiency [GFLOPS@16b/cycle]	39	177

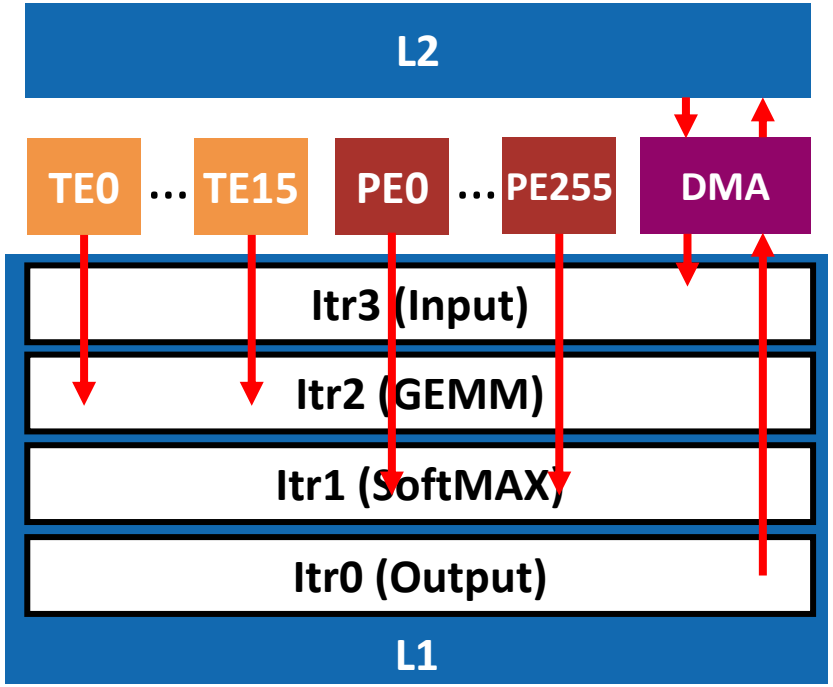
- 6x Throughput because of 2x more FMAs and 3x better utilizazion 89% vs 30%
- 4.4x Energy efficiency, 4.5x Area&Energy Efficiency

Performance of other kernels & Pipelining options

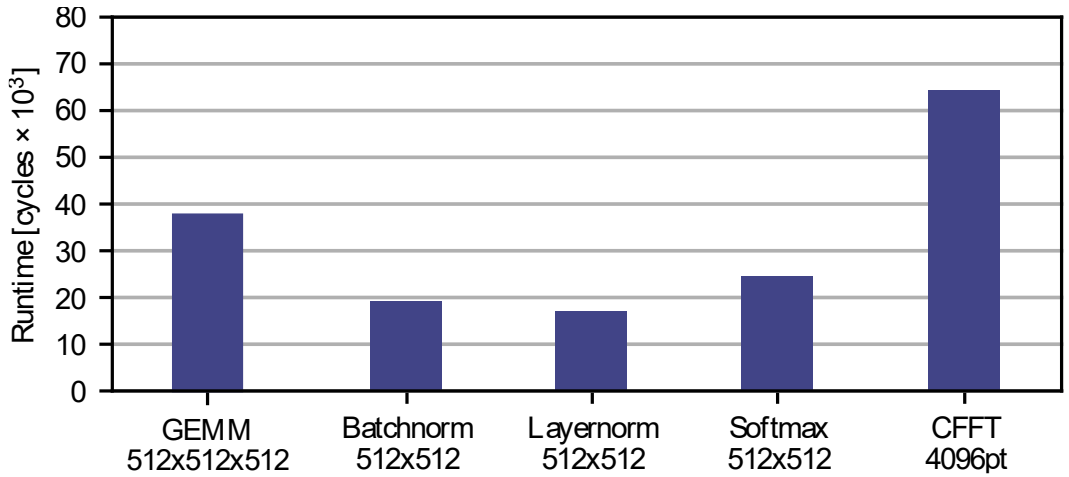
- **PEs enable concurrent execution of activation functions**
- **I tested the performance on:**
  - Classical signal processing kernels (4096pt-CFFT)
  - Non-linear activation functions (Layernorm, BatchNorm, SoftMax)

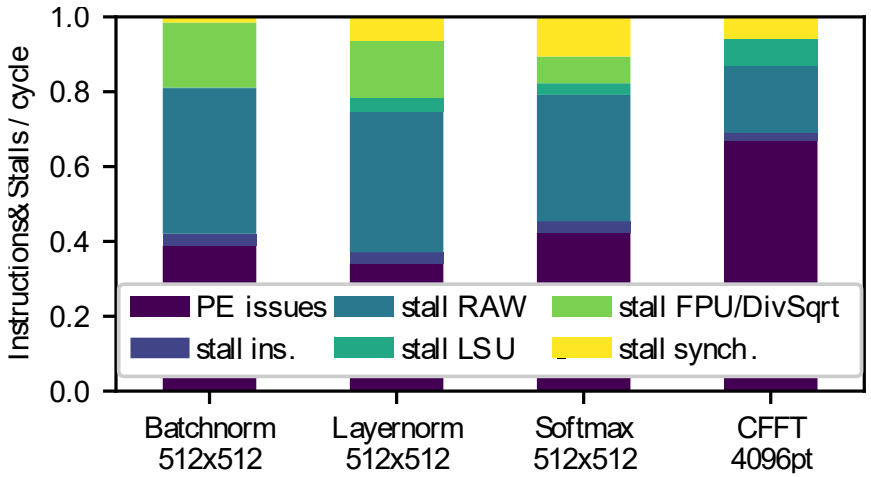
Steps of computation:

1) L1 holds the data for 4 iterations
2) DMA transfers in data for itr3
3) TEs execute GEMM on itr2
4) PEs compute SoftMax on itr1
5) DMA transfers result (itr0)



Performance of other kernels & Pipelining options





- **The previous parallelization scheme is possible because**
  1. PEs can execute the activation concurrently
  2. The runtime of activations on PEs is lower than the runtime of GEMM on the TEs
- **Additionally we can keep executing classical signal-processing tasks (CFFT)**

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